GEORGE LARSON

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SUMMARY

Updated 2026-09 Director of Technology with 25+ years building and operating software, infrastructure, and manufacturing systems. I have led teams at Sony Electronics, scaled a medical billing startup to national reach, programmed PLCs on production lines, and directed technology for a mid-size manufacturer. I do Linux platforms, DevOps automation, ecommerce, and security programs. I stay close to the keyboard even when the job is strategic. I build AI tooling that actually ships: multi-agent infrastructure, MCP servers, model-routing pipelines, and a personal review methodology with twenty-nine fixes merged upstream across a dozen projects, twenty-one of them in the OpenHands ecosystem, which named me an Active Contributor in August 2026.

CORE STRENGTHS

Software Development and Engineering

- Full stack delivery across PHP, JavaScript, SQL, Python, Go, Rust, C#, TypeScript, C++, with Perl as a long-standing second string.
- Code review, automated testing, and performance tuning for legacy and greenfield systems.

DevOps and Infrastructure Automation

- CI and CD design, infrastructure as code, containers (Docker, Podman), and repeatable releases on self-hosted runners.
- Observability, incident response, high-availability design (replication, backups, CDN), and pragmatic SLO tracking for hybrid environments.

Technology Leadership and Strategy

- Strategic roadmaps that connect executive goals to team-level delivery.
- Calm stewardship for blended internal, offshore, and vendor teams.

Security and Privacy

- Network hardening, IAM, HostedScan/ZAP/Nikto assessment loops, and zero-trust rollouts.
- Regulated-data guardrails and tabletop response drills.

Data Engineering and Automation

- OCR digitization, ETL pipelines, semantic search, and knowledge-base design.
- Workflow automation using scripting, open models, and RAG pipelines where the payoff is real.

OPEN SOURCE PROJECTS

Recent Upstream Contributions Twenty-nine merged pull requests across twelve projects, twenty-six of them since June 2026. Every entry links a merged PR at georgelarson.me/contributions.

Merged, OpenHands ecosystem (21; named an Active Contributor, August 2026)

- OpenHands app (12): context-window usage meter, usage drawer, and manual compaction (#16311, #16534); the Linux desktop installer build (#16656); SIGHUP handling across all three dev launchers (#16239, #16279, #16280); the ACP credentials banner driven by the secret store (#16145); cloud provider picker following the pagination cursor (#16758); the LLM-switching toggle and active-model stamp (#16441, #16439); mock-LLM end-to-end shutdown (#16152); runtime state ignored at the .tmp/ root (#16272).
- Software-agent SDK, docs, and clients (9): boundary tests for events_to_messages that exposed and fixed a live defect dropping reasoning items (#4526); POST /file/create_directory on the file router (#4482); send_message skill-activation wiring (#4536); Kimi Code as a first-class Moonshot provider across the SDK and Agent Canvas (#4150, #1844); the issue-triage and ready-for-dev gate page (docs #750); the release-gate permission fix, the first fork PR merged in the typescript-client repo (#364); the Agent Canvas workspace-folder dropdown and acp-docker example (#1368, #1434).

Merged, beyond OpenHands (8)

- kputnam/stupidedi #288, #289: re-enabled the X12 4010 PO850, AR943, and RE944 EDI transaction sets, whose autoload had been off after the segment definitions outgrew their element lists. Merged July 2026.
- juspay/neurolink #1134: moved system messages to the AI SDK generateText system option; shipped in v9.82.0 on npm. Merged July 2026.
- ferriskey/ferriskey #1133: added the RFC 8693 TokenExchange grant type and its domain value objects to the Rust auth server. Merged July 2026.
- mvanhorn/last30days-skill #564: pass(1) credential source, a Linux/Unix analog of the macOS Keychain integration. Merged June 2026.
- sst/models.dev #475: added Claude Opus 4.5 to the model catalog. Merged Dec 2025.
- sst/opencode #4748: documented the Venice.ai provider for the opencode coding agent. Merged Nov 2025.
- WPBuddy/largo #164: the long end of the track record, 2014.

In review

- OpenHands/software-agent-sdk #4775 and docs #771: preserve structured MCP result data (structuredContent and result _meta) through the observation boundary, the contract protocol-aware clients build on.
- OpenHands/software-agent-sdk #4623: an offline prompt-composition report that breaks each LLM call's prompt into system, tool-schema, history, and latest-message buckets.
- livekit/agents #6280: duplicate same-target handoff tool call no longer reads as ambiguous and aborts the agent switch.
- allenai/open-instruct #1751: a –launcher flag so non-Ai2 users can run the training launcher locally without a Beaker cluster account.
- stryker-mutator/stryker-net #3694: extends the array-creation mutator to implicit array creation.
- mitmproxy/mitmproxy #8199, #8200: Flow.kill() resume-event and KillInjected fixes so teardown closes the client connection and disconnects in-transit flows.
- charmbracelet/crush #2749, #2750: LSP score-based client selection and single-file-server handling so single-file servers start without root markers.

Fixed upstream (idea shipped under someone else's commit)

- n8n-io/n8n #29516: per-runner consecutive-timeout counter for stuck task-runner channel detection and recovery; reduced detection latency from hours to ~6s. A maintainer re-implemented the idea on current master and credited the diagnosis by name (#35456); shipped in the 2.33 and 2.34 release lines.
- OpenHands #16415 and #16443, charmbracelet/catwalk #112: the same problems later fixed by others' merged PRs.

deep-research-privacy (45 stars, TypeScript)

- Standalone privacy-first AI research tool: Brave Search, zero data collection, parallel queries with cited output, and an iterative search-and-

synthesis pipeline.

Venice AI SDKs (2025 to 2026, archived)

- Published TypeScript and PHP SDKs (venice-dev-tools, venice-ai-php) for the Venice API. Dated craft work; repositories archived 2026.

LLM Showdown (2026)

- llm-showdown-data. Companion dataset for the blog series benchmarking 15+ free coding models on a $25 VPS using opencode and a ralph-wiggum-style iterative loop.
- Designed and runs the evaluation rig; publishes per-round results comparing model behavior on real coding tasks.

Fracture (2 stars, TypeScript), Modernization Case Study (2026)

- Modernizes Mozilla's 2012 BrowserQuest into a contemporary multiplayer architecture with eight integrated AI services: RAG, semantic search, agent-driven triage, and content moderation.
- Live demo at fracture.georgelarson.me. The case study itself is the deliverable: how to take a frozen reference codebase from a major vendor and bring it forward without rewriting from scratch.

nullclaw + ironclaw, Agent Mesh (Zig + Rust, 2026)

- nullclaw: public-facing AI concierge ("Nully") on a hardened perimeter box. Tiered inference (Haiku for conversational triage, Sonnet for code analysis). Ran a full self-hosted comms stack (Ergo IRC, gamja web client, Cloudflare-proxied WebSocket, Let's Encrypt TLS) before transitioning to a static-site origin.
- ironclaw: private personal agent on a separate system with distinct security boundaries; agent-to-agent routing ran between the two until the daemon stack was retired June 2026.
- Extended into a multi-node Agent Zero mesh with A2A routing, dedicated chat and utility models, and per-node project isolation.

netgrep (Rust), Creator and Maintainer (2026 to Present)

- Modern replacement for ngrep: live packet capture and pcap analysis with TCP stream reassembly, TLS 1.2/1.3 decryption, HTTP/2 and DNS parsing, and an interactive terminal UI.
- Maintains ngrep-compatible CLI flags; outputs colored text, JSON, hex dump, or pcap.

termshark (Go), Contributor and Maintainer (2026 to Present)

- Modernized fork of termshark, a terminal UI for Wireshark/tshark.
- Refactored architecture, improved test coverage, updated to Go 1.22+ idioms, and added an experimental browser-based web interface.

wiregraph (Rust, 2026 to Present)

- GPU-rendered network traffic visualizer. Hosts as nodes, flows as edges, real-time topology view of a live capture session.

rust-connect (Rust, 2026 to Present)

- Desktop-agnostic KDE Connect reimplementation, API-first, pairing with the stock KDE Connect Android app with no KDE or GNOME dependencies. Feature-complete against the reference implementations; v0.1.0 released with deb packages.

pgh-hospital-prices (Go, 2026)

- Compares what Pittsburgh hospitals charge for the same procedure, built over the federal hospital price-transparency files: corpus, query grammar, and a public dashboard at prices.georgelarson.me.

Agent Canvas extensions (JavaScript, OpenHands, 2026)

- Two Canvas Extensions for the OpenHands agent stack: agent-canvas-subagent-monitor watches parallel task-tool subagent calls in a live conversation; agent-canvas-usage-explorer makes recorded LLM usage inspectable across runs.

WORK EXPERIENCE

Director of Technology, FM Expressions (2018 to April 2026)

- Led technology strategy for a mid-size manufacturing company serving ecommerce channels.
- Absorbed the infrastructure team and its full stack when the role was eliminated during a downturn, running technology and infrastructure together with no added headcount.
- Shipped production AI inside the business from 2023: a customer-facing DALL-E design assistant for custom heat transfers, an OpenAI/Pythagora code-generation tool on MongoDB, an AI knowledge platform (wiki.fmx.ink), a customer-service AI/BI integration, and a Python/FastAPI templating system bridging the Fanatics and VivX vendor pipelines.
- Improved database performance by 93 percent while driving critical systems to 99 percent uptime; cut MySQL slow-query volume by 96 percent to accelerate incident response.
- Built a knowledge management platform (Wiki.js, Gitea, Qdrant, ETL pipelines) that captures production logs, transcripts, and operator notes so teams can search, surface, and act on institutional knowledge in real time.
- Resolved conflicting manufacturing documentation, created a style guide and glossary, and instituted monthly and quarterly review cycles to keep safety and operations guidance consistent.
- Reduced a 1.03 GB legacy software repository (4,656 binary artifacts) to a clean, LFS-ready codebase, eliminating clone failures and enabling sustainable development practices.
- Hardened hybrid infrastructure with Authentik SSO, HashiCorp Vault, automated TLS renewal, firewall hardening (iptables, UFW, CrowdSec, fail2ban), DMARC/SPF/DKIM enforcement, HostedScan/OWASP ZAP sweeps across 36 public assets, and Red Sift OnDMARC (p=reject, 1.5M emails/month at 99.48% compliance) to keep critical/high findings at zero; closed 33 issues in the 60 days prior to separation.
- Ran the security program's people-and-email layer: Mimecast email security, a KnowBe4 security-awareness and phishing-simulation program, and live response to an active production compromise.
- Ran the collaboration and operations core: self-hosted GitLab CI/CD, Mattermost team comms (retired 2026), Redmine project tracking, a Zammad helpdesk clearing ~200 tickets/month with 68% closed inside 24 hours, and Uptime Kuma monitoring feeding alert channels.
- Maintained high-availability platforms across a heterogeneous Windows-and-Linux estate: a high-availability MySQL cluster and MSSQL, VMware ESXi with Datto backups, on-prem Active Directory with Entra ID hybrid identity (Group Policy, PowerShell), multi-vendor firewalls (Cisco, SonicWall, pfSense), Cisco Meraki switching, OpenVPN, and a staged CDN migration from self-hosted edges to Contabo and Cloudflare; led countless OS and database migrations.
- Built the Mashy internal dashboard plus tooling such as Psono vault, PrivateBin, phpMyAdmin/Adminer, DBeaver, knowledge mining utilities, and status pages so teams access credentials, data stores, and observability in one click.
- Drove adoption metrics for the knowledge stack to quantify time-to-answer gains for production and customer operations.
- Captured Meteor print-head PCAPs and tuned Nikto, Metasploit, and custom fuzz rigs to secure industrial protocols while leading the move from VPN toward zero-trust access.
- Implemented Agile delivery, ticketing, and analytics to align product, operations, and finance on throughput, defect, and margin targets.
- Developed and operated Dashboard, FM's internal proprietary system for everything: orders, kiosk, production, prepress, preflight, planner,

users, and servers, with importers feeding it from Impress, Boxseat, and Printro. PHP and MySQL with JavaScript and jQuery throughout.
- Took over Printro, FM's own C# SaaS Digital Art Management platform for ordering heat transfers, and ran it: the API, the dashboard, the Azure infrastructure, CI/CD, and the roadmap. Its order EDI into Dashboard was a Node.js service.
- Integrated and operated the two MIS/ERP systems the business ran on, EFI Pace and Aptean Impress: Power BI ETL against the Pace database, the Impress ProcessXML integration and order importer, staging environments, and their hosting and migrations.
- Wrote the C# integration to VivX, the templated-art engine on a Windows box in Azure that takes a customer's name, sets it on the jersey in the right font, and builds a print-ready PDF for the heat transfer.
- Built and operated integrations across the print and fulfillment vendor stack (EFI Pace, Aptean Impress, VivX, Fanatics, Alphabroder, SignTronics) plus ecommerce platforms (Shopify, Magento) and marketing tooling (HubSpot, Power BI), so the production floor, order pipeline, and analytics layer share a coherent data model.
- Administered the prepress production stack (Esko, Prinergy, RIP workflows) and 3PL logistics integrations that kept the manufacturing floor running.
- Built financial-modeling tools using AI/ML to analyze company performance and forecast scenarios; presented findings directly to executive leadership to inform strategic decisions during a difficult business period.

Senior Software Engineer and Manager, Universal SmartComp (2009 to 2016)
- Helped scale the company from a 50 person startup to a national leader in medical billing (Inc 5000, ranked #34 in Health and #503 overall).
- Designed the EDI framework, ACH transmission, ERP integration, payer portal, Gold Standard Review, and recovery analytics layers of the platform.
- Led internal and offshore teams across development and Linux infrastructure.
- Acted as security response lead, HIPAA/HITECH SME, and Linux SME for the medical billing platform; migrated project management to Redmine with custom plugins.
- Deployed VMware ESXi covering web, database, DHCP, DNS, and ERP servers across the platform.
- Built OCR bill entry validation that reduced errors and sped turnaround.

Software Engineer and Team Manager, Sony Electronics (2004 to 2009)
- Joined a service operation that was effectively bare shelves at start; directed software development, logistics, and technical operations for the Pittsburgh Customer Satisfaction Center from there.
- Designed and ran the TiVo firmware upgrade station for the United States market.
- Built hardware testing suites for CD drives, set-top boxes, PCBs, televisions, cameras, and MP3 players.
- Created an AS/400 parts ordering interface that saved millions in operational spend.
- Front-end software work delivered roughly 30 percent productivity gain and 12 percent expense reduction.
- Managed more than 50 combined technical and non-technical staff, including roughly 30 contracted personnel.
- Developed a VT-52 terminal emulation system used for inventory management.

Maintenance and Automation Technician, T. Marzetti Company (2000 to 2004)
- Led second shift maintenance for robotics, conveyors, and packaging machinery.
- Programmed PLCs and Adept V++ robotics in-house, tuned sensors, actuators, and industrial controls on the production line, and kept robotics development internal to limit vendor cost.

Earlier Career Highlights (1990s to 2000s)
- Software developer and webmaster for non-profit organizations.
- Engineer for The Sounding Board, a psychotherapeutic BBS in Columbus, Ohio; built custom C++ modules for Major BBS and kept one of the first internet-connected BBS systems alive.
- Contract developer for Daifuku USA working on material handling support systems.
- Paid programming and database work 1994 to 2000 via Pacific Judgment Recovery (Brice, OH), AGL Personnel (Grove City, OH), and a Remedy Staffing CNC contract.
- Two-time Business Professionals of America Chapter 13 president; state and regional wins in Computer Programming, Team Mathematics, and Extemporaneous Speech.

INDEPENDENT WORK

- Rebuilt a teacher's archive site (2026): migrated a WordPress site to a static build served from the owner's own Cloudflare account; downloaded and transcribed 645 recordings on a home GPU with cloud overflow; published 681 searchable talk pages with browser-side full-text search, a calendar feed, and a no-script newsletter signup; preserved all original URLs; handed over source, transcripts, and how-to guides so the owner maintains it herself. Writeup at georgelarson.me/writing/2026-10-16-sailorbob.

VOLUNTEER WORK

- Systems administrator for a volunteer Linux community, digital radio station, and open shell environment.
- Long-term support for animal shelters, including events, outreach, and facilities work.
- OCR and digitization project contributor for historical legal documents.

TECHNICAL SKILLS

Languages: PHP, JavaScript, SQL, Bash, Python, Go, Rust, TypeScript, C#, Perl, Zig, C++ Web and Data: MySQL, MSSQL, PostgreSQL, WordPress, Redmine, Gitea, Wiki.js Infrastructure: Linux, Active Directory, Entra ID, Group Policy, PowerShell, Apache, Nginx, Caddy, IIS, VMware ESXi, Datto backups, Cisco firewalls, Cisco Meraki, SonicWall, pfSense, OpenVPN, CrowdSec, Let's Encrypt/ACME, Cloudflare CDN Cloud: AWS, Azure, Linode, Rackspace, Google Cloud, Proxmox Security and Identity: Authentik, HashiCorp Vault, Psono, CrowdSec, HostedScan, OWASP ZAP, Nikto, Metasploit, SSL/TLS, iptables, UFW, fail2ban, DMARC/SPF/DKIM, Mimecast, KnowBe4, DefectDojo, OpenVAS, zero-trust network design, incident response, CCPA/GDPR compliance work Collaboration: Agile, Scrum, Kanban, Redmine, Zammad, GitLab, CryptPad Mail Systems: iRedMail, Postfix, Dovecot Commerce: Shopify, custom ecommerce platforms, payment integrations Data and Automation: RAG design, OCR pipelines, semantic search, ETL, knowledge base design AI Agent Infrastructure: Agent Zero (A2A protocol, multi-node mesh, project isolation), Model Context Protocol (MCP server design), Composio toolkit integration, Venice AI SDK ecosystem (TypeScript/PHP SDKs), LiteLLM, multi-provider model routing across 12 authenticated providers (Anthropic, OpenAI, Z.AI GLM, MiniMax, Moonshot/Kimi, Airouter, Groq, Cerebras, Featherless, OpenRouter, NVIDIA, Mistral), tiered inference (utility/chat split), prompt caching, agent-driven CI and content pipelines, RAG and semantic search, agent security boundaries, scheduled autonomous tasks. Operational fluency across ten coding-agent CLIs (Claude Code, opencode, gemini-cli, Codex CLI, Kimi Code, OpenHands, Aider, Octofriend, Crush, letta-code) used as a benchmark and production stack. Systems and Networking: TCP/IP analysis, packet capture, PLC programming, industrial controls, prepress production (Esko, Prinergy, RIP workflows), print MIS and ERP (EFI Pace, Aptean Impress), IRC server administration Monitoring & Dashboards: Uptime Kuma, HostedScan,

Red Sift OnDMARC, Dashy internal portal

EDUCATION AND PROFESSIONAL DEVELOPMENT

- Self-taught from C++ books as a teenager; built BBS modules in production before attending college.
- Continuous learner across hundreds of online courses, lectures, and certifications (see LinkedIn for details).
- Currently studying: Stanford CS336, Language Modeling from Scratch (2025).
- Additional coursework spanning systems engineering, network security, database internals, and cloud architecture through platforms including Coursera, MIT OpenCourseWare, and vendor certification programs.